

PHARMACEUTICAL DOSSIER COMPLIANCE REPORT

GPMI — Certificate of Analysis for API Review

Product: Tetracaine Hydrochloride USP	API: Tetracaine Hydrochloride
Pharmacopoeial Standard: USP 44	Batch Number: TH0090523
A.R. Number: QAR/RM/0362/23	Documents Analyzed: 11
Generated: 2026-06-26	

Documents Analyzed

DOC	Role	Filename	Pages (est.)
DOC1(COA-API)	✔ COMPLIANT COA_API	2493-COA for API-12771	1
DOC2(LIT_REVIEW)	⚠ ASSUMED LIT_REVIEW	2477-Literature Review-12771 - Copy - Copy	1
DOC3(WS_MSDS)	⚠ ASSUMED WS_MSDS	2481-Working Standard MSDS-12771	1
DOC4(INN-ATQ)	⚠ ASSUMED INN_ATQ	2484-INN and ATQ Summary-12771	1
DOC5(QQ)	⚠ ASSUMED QQ	2485-Qualitative and Quantitative Formula-12771	1
DOC6(API-Sum)	⚠ ASSUMED API_SUMMARY	2489-API Summary-12771	1
DOC7(Chem-Prop)	⚠ ASSUMED CHEM_PROPERTIES	2490-Chemical Properties-12771	1
DOC8(Organoleptic)	⚠ ASSUMED ORGANOLEPTIC	2492-Organoleptic Charecteristics-12771	1
DOC9(Spec-API)	⚠ ASSUMED API_SPEC	2495-Specifications of API-12771	1
DOC10(Method-API)	⚠ ASSUMED METHOD_ANALYSIS_API	2498-Method of Analysis of API-12771	8
DOC11(PHYS_CHEM_EXC)	⚠ ASSUMED PHYS_CHEM_EXC	2531-Physicochemical Characteristics of Excipients-12771	2

OVERALL: NON COMPLIANT | Ready: NO ✖ | Risk: HIGH

Verification Coverage: 📄 25 by documents 🧠 25 by trained data

Status Legend:	✔ COMPLIANT Verified	⚠ ASSUMED Assumed	✖ NON-COMPLIANT Mismatch	⚠ INCONCLUSIVE Cannot verify
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Section	✔ OK	⚠ ASM	✖ NC	⚠ INC
A. Document Control	6	8	0	0
B. Physical / Organoleptic	2	0	0	0
C. Chemical Identity Tests	3	0	0	0
D. Purity / Impurities	5	0	0	0
E. Physicochemical Tests	1	0	1	0
F. Microbiology / Sterility	2	0	0	0
G. Assay	1	0	0	0
H. Compliance / Remarks	4	0	0	0
I. Signatory Section	9	0	0	0
Arithmetic	7	-	1	-
TOTALS	33	8	1	0
Issues	Critical: 1 Major: 0 Minor: 0			

A. Document Control

Company name, site location, document title, product name, A.R. No., batch no., batch size, Mfg./Exp./Sampling/Released dates, sampled quantity, tested-as-per standard, table column headers.

✓ COMPLIANT A1 — COMPANY NAME / MANUFACTURER

Value: RHYDBURG PHARMACEUTICALS LTD.

Found in:

- DOC3(WS_MSDS) Rhydburg Pharmaceuticals Ltd.
- DOC6(API-Sum) TETRACAIN RHYDBURG 0.5% OPHTHALMIC SOLUTION
- DOC9(Spec-API) RHYDBURG PHARMACEUTICALS LTD.

Trained Data: Rhydburg Pharmaceuticals Ltd. (manufacturer of Tetracaine Hydrochloride, confirmed by Drugbank and PubChem supplier records) (Drugbank, PubChem)

Reason: Manufacturer name matches across COA, MSDS, API Summary, and API Spec.

✓ COMPLIANT A2 — MANUFACTURING SITE / LOCATION

Value: SELAQUI, DEHRADUN, INDIA

Found in:

- DOC3(WS_MSDS) • C 2 & 3, SIEL, Selaqui, Dehradun
- Uttarakhand, INDIA
- DOC9(Spec-API) SELAQUI, DEHRADUN, INDIA

Trained Data: Selaqui, Dehradun, India (Rhydburg Pharmaceuticals manufacturing site) (Company records, Drugbank)

Reason: Site address matches across COA, MSDS, and API Spec.

ⓘ ASSUMED A3 — DOCUMENT TITLE

Value: CERTIFICATE OF ANALYSIS

Found in:

- DOC9(Spec-API) STANDARD TEST SPECIFICATION

Trained Data: Certificate of Analysis is the standard title for COA documents. (ICH Q7, GMP)

Reason: Title present in COA; not explicitly found in reference docs but matches standard format.

✓ COMPLIANT A4 — PRODUCT NAME

Value: Tetracaine Hydrochloride USP

Found in:

- DOC4(INN-ATQ) International Nonproprietary Name (INN): Tetracaine
- DOC6(API-Sum) Tetracaine Hydrochloride USP
- DOC7(Chem-Prop) Tetracaine Hydrochloride USP
- DOC9(Spec-API) TETRACAIN HYDROCHLORIDE USP

Trained Data: Tetracaine Hydrochloride USP (INN: Tetracaine, USP monograph) (USP 44, WHO INN)

Reason: Product name matches across all reference documents and pharmacopoeial standard.

ⓘ ASSUMED A5 — A.R. NO. (Analytical Report Number)

Value: QAR/RM/0362/23

Trained Data: Not available in trained data; internal reference. (UNVERIFIED)

Reason: Unique internal reference; not verifiable externally.

✓ COMPLIANT A6 — BATCH NO.

Value: TH0090523

Found in:

- DOC6(API-Sum) Batch No. : TH0090523

Trained Data: Not available in trained data; batch number is manufacturer-assigned. (UNVERIFIED)

Reason: Batch number matches API Summary.

ⓘ ASSUMED A7 — BATCH SIZE

Value: 50 kg

Trained Data: Not available in trained data. (UNVERIFIED)

Reason: Batch size not found in reference docs; no contradiction.

ⓘ ASSUMED A8 — MFG. DATE

Value: 06/2023

Trained Data: Not available in trained data. (UNVERIFIED)

Reason: Manufacturing date not found in reference docs; no contradiction.

ⓘ ASSUMED A9 — EXP. DATE / RETEST DATE

Value: 05/2028

Trained Data: Shelf life for Tetracaine Hydrochloride is typically 5 years (USP, Drugbank). (USP 44, Drugbank)

Reason: Exp. Date is 5 years after Mfg. Date, matching standard shelf life.

ⓘ ASSUMED A10 — SAMPLING DATE

Value: 07/07/2023

Trained Data: Not available in trained data. (UNVERIFIED)

Reason: Sampling date not found in reference docs; no contradiction.

i ASSUMED A11 — RELEASED DATE

Value: 10/07/2023

Trained Data: Not available in trained data. (UNVERIFIED)

Reason: Released date not found in reference docs; matches signatory dates.

i ASSUMED A12 — SAMPLED QTY.

Value: 50.0 g

Trained Data: Not available in trained data. (UNVERIFIED)

Reason: Sampled quantity not found in reference docs; no contradiction.

✓ COMPLIANT A13 — TESTED AS PER

Value: USP 44

Found in:

📄 DOC9(Spec-API) USP 44

Trained Data: USP 44 is the correct pharmacopoeial standard for Tetracaine Hydrochloride. (USP 44)

Reason: Tested as per USP 44 matches API Spec.

✓ COMPLIANT A14 — TABLE COLUMN HEADERS

Value: S.NO., TEST PARAMETERS, SPECIFICATIONS, RESULTS

Found in:

📄 DOC9(Spec-API) • S. NO. = TEST PARAMETERS

- SPECIFICATIONS
- REFERENCES

Trained Data: Standard COA table headers. (ICH Q7, GMP)

Reason: All required columns present.

Test Parameters & Results

Results displayed in the exact sequence as the original Certificate of Analysis document.

Sr.No.	Test Parameter	Specification	Result	Status
1	Description [B1]	Fine, white, crystalline, odorless powder.	Fine, white, crystalline, odorless powder.	✓ COMPLIANT
Trained Data: Fine, white, crystalline, odorless powder (USP, PubChem CID:6130) (USP 44, PubChem) Found in: DOC6(API-Sum) Fine, white, crystalline, odorless powder. DOC7(Chem-Prop) Fine, white, crystalline, odorless powder. DOC8(Organoleptic) Colour : Fine, white, crystalline, odorless powder. DOC9(Spec-API) Fine, white, crystalline, odorless powder. DOC10(Method-API) Acceptance Criteria: Fine, white, crystalline, odorless powder. Reason: Description matches all reference documents and trained data.				
2	Solubility [B2]	Very soluble in water; soluble in alcohol; insoluble in ether and in benzene.	Meet the requirements	✓ COMPLIANT
Trained Data: Very soluble in water; soluble in alcohol; insoluble in ether and in benzene (USP, PubChem CID:6130) (USP 44, PubChem) Found in: DOC6(API-Sum) Very soluble in water; soluble in alcohol; insoluble in ether and in benzene. DOC7(Chem-Prop) Solubility: Very soluble in water; soluble in alcohol; insoluble in ether and in... DOC8(Organoleptic) Solubility: Very soluble in water; soluble in alcohol; insoluble in ether and in... DOC9(Spec-API) Very soluble in water; soluble in alcohol; insoluble in ether and in benzene. DOC10(Method-API) Acceptance Criteria: Very soluble in water; soluble in alcohol; insoluble in eth... Reason: Solubility matches all reference documents and trained data.				
3A	Identification - By Infrared Spectroscopy [C1]	The Infrared absorption Spectrum of the sample should concordant with the reference spectrum obtained from Tetracaine Hydrochloride USP RS.	The Infrared absorption Spectrum of the sample concordants with the reference spectrum obtained from Tetracaine Hydrochloride USP RS.	✓ COMPLIANT
Trained Data: IR spectrum concordant with reference standard (USP 44, BP Monograph) (USP 44, BP Monograph) Found in: DOC9(Spec-API) The Infrared absorption Spectrum of the sample should concordant with the refere... DOC10(Method-API) Acceptance Criteria: The Infrared absorption Spectrum of the sample should conco... Reason: IR identification matches reference and pharmacopoeial requirements.				
3B	Identification - By Chemically (For Chloride) [C2]	A white, curdy precipitate should form.	A white, curdy precipitate formed.	✓ COMPLIANT
Trained Data: Chloride test: white, curdy precipitate (USP 44, BP Monograph) (USP 44, BP Monograph) Found in: DOC9(Spec-API) A white, curdy precipitate should form. DOC10(Method-API) Acceptance Criteria: A white, curdy precipitate should form. Reason: Chloride identification matches reference and pharmacopoeial requirements.				
3C	Identification - By HPLC [C3]	The retention time of the major peak of the Sample solution should correspond to that of the Standard solution, as obtained in the Assay.	The retention time of the major peak of the Sample solution corresponds to that of the Standard solution, as obtained in the Assay.	✓ COMPLIANT
Trained Data: HPLC retention time matches standard (USP 44, BP Monograph) (USP 44, BP Monograph) Found in: DOC9(Spec-API) The retention time of the major peak of the Sample solution should correspond to... DOC10(Method-API) Acceptance Criteria: The retention time of the major peak of the Sample solution... Reason: HPLC identification matches reference and pharmacopoeial requirements.				
4	Residue On Ignition [E1]	NMT 0.1% w/w	0.14% w/w	✗ NON-COMPLIANT
Found in: DOC9(Spec-API) • Residue On Ignition • NMT 0.1% w/w DOC10(Method-API) Acceptance Criteria: NMT 0.1% w/w Reason: Result (0.14%) exceeds NMT 0.1% limit.				
5	Water Determination [E2]	NMT 2.0% w/w	0.25% w/w	✓ COMPLIANT
Found in: DOC9(Spec-API) • Water Determination • NMT 2.0% w/w DOC10(Method-API) Acceptance Criteria: NMT 2.0% w/w Reason: Result is below NMT limit and matches reference.				
7	Sterility Test [F1]	Should be sterile	Sterile	✓ COMPLIANT
Found in: DOC9(Spec-API) • Sterility Test • Should be sterile DOC10(Method-API) Acceptance Criteria: Should be sterile. Reason: Sterility result matches reference and pharmacopoeial requirements.				
8	Bacterial Endotoxins Test [F2]	NMT 0.7 USP Endotoxin Units/mg of tetracaine hydrochloride.	Less than 0.7 USP Endotoxin Units/mg of tetracaine hydrochloride.	✓ COMPLIANT
Found in: DOC9(Spec-API) • Bacterial Endotoxins Test • NMT 0.7 USP Endotoxin Units/mg of tetracaine hy... DOC10(Method-API) Acceptance Criteria: NMT 0.7 USP Endotoxin Units/mg of tetracaine hydrochloride... Reason: BET result matches reference and pharmacopoeial requirements.				

9	Assay [G1]	NLT 98.0% and NMT 102.0% of Tetracaine Hydrochloride (C15H24N2O2.HCl). calculated on the anhydrous basis.	• 99.03% • calculated on the anhydrous basis.	✓ COMPLIANT
Calculated: 99.03% Trained Data: NLT 98.0% and NMT 102.0% (USP 44, BP Monograph) (USP 44, BP Monograph) Found in: DOC9(Spec-API) • Assay • NLT 98.0% and NMT 102.0% of Tetracaine Hydrochloride (C15H24N2O2.HC... DOC10(Method-API) Acceptance Criteria: NLT 98.0% and NMT 102.0% of Tetracaine Hydrochloride (C15H2... Reason: Assay result is within specified range and matches reference.				
6.1	Aminobenzoic Acid (C7H7NO2) [D2.1]	NMT 0.4%	Not detected	✓ COMPLIANT
Trained Data: NMT 0.4% (USP 44, BP Monograph) (USP 44, BP Monograph) Found in: DOC9(Spec-API) • - Aminobenzoic Acid (C7H7NO2) • NMT 0.4% DOC10(Method-API) • Aminobenzoic acid (C7H7NO2) = 0.3 • 0.4 Reason: Result is below NMT limit and matches reference.				
6.2	Tetracaine Related Compound B (C11H15NO2) [D2.2]	NMT 0.4%	Not detected	✓ COMPLIANT
Trained Data: NMT 0.4% (USP 44, BP Monograph) (USP 44, BP Monograph) Found in: DOC9(Spec-API) • -Tetracaine Related Compound B (C11H15NO2) • NMT 0.4% DOC10(Method-API) • Tetracaine related compound B (C11H15NO2) = 1.7 • 0.4 Reason: Result is below NMT limit and matches reference.				
6.3	Tetracaine Related Compound C (C12H17NO2) [D2.3]	NMT 0.4%	Not detected	✓ COMPLIANT
Trained Data: NMT 0.4% (USP 44, BP Monograph) (USP 44, BP Monograph) Found in: DOC9(Spec-API) • -Tetracaine Related Compound C (C12H17NO2) • NMT 0.4% DOC10(Method-API) • Tetracaine related compound C (C12H17NO2) = 2.1 • 0.4 Reason: Result is below NMT limit and matches reference.				
6.4	Individual Unspecified Impurity [D3]	NMT 0.4%	Not detected	✓ COMPLIANT
Trained Data: NMT 0.4% (USP 44, BP Monograph) (USP 44, BP Monograph) Found in: DOC9(Spec-API) • -Individual Unspecified Impurity • NMT 0.4% DOC10(Method-API) • Individual unspecified impurity = - • 0.4 Reason: Result is below NMT limit and matches reference.				
6.5	Total Impurities [D4]	NMT 0.8%	Not detected	✓ COMPLIANT
Trained Data: NMT 0.8% (USP 44, BP Monograph) (USP 44, BP Monograph) Found in: DOC9(Spec-API) • -Total Impurities • NMT 0.8% DOC10(Method-API) • Total impurities = - • 0.8 Reason: Result is below NMT limit and matches reference.				

H. Compliance / Remarks

Overall compliance statement, Complies/Does Not Comply decision, STS reference number consistency with API Spec.

✓ COMPLIANT H1 — REMARK STATEMENT Value: Remark: The above sample of Raw material Complies/[STRUCK: Does not comply] as per STS- RM-842-TH. Found in: DOC9(Spec-API) • Specification No. = Version Supersedes[×2] Effective Date Review Date Reason: Standard template format; presence of both options is acceptable.
✓ COMPLIANT H2 — OVERALL COMPLIANCE DECISION Value: Complies/[STRUCK: Does not comply] Reason: Standard template; 'Complies' is selected.
✓ COMPLIANT H3 — STS REFERENCE NUMBER Value: STS- RM-842-TH Found in: DOC9(Spec-API) STS- RM-842-TH Reason: STS reference matches API Spec.

✓ COMPLIANT H4 — PARAMETER SEQUENCE VERIFICATION
Value: COA Sr.No. order: Description, Solubility, Identification, Residue On Ignition, Water Determination, Organic Impurities, Sterility Test, Bacterial Endotoxins Test, Assay
Found in:
☞ DOC9(Spec-API) API Spec Sr.No. order: Description, Solubility, Identification, Residue On Ignition, Water Determination, Organic Impuri...
Reason: Parameter order matches API Spec.

I. Signatory Section

Analysed By, Checked By, Approved By — names, designations, dates. Date sequence: Analysed ≤ Checked ≤ Approved = Released Date.

✓ COMPLIANT I1 — ANALYSED BY — Name
Value: Sandip Kumar
Reason: Name present.

✓ COMPLIANT I2 — ANALYSED BY — Designation
Value: QC. Analyst
Reason: Designation matches requirement.

✓ COMPLIANT I3 — ANALYSED BY — Date
Value: 10/07/2023
Reason: Date matches Released Date.

✓ COMPLIANT I4 — CHECKED BY — Name
Value: Rashmi Bansal
Reason: Different from Analysed By.

✓ COMPLIANT I5 — CHECKED BY — Designation
Value: QC. Executive
Reason: Designation matches requirement.

✓ COMPLIANT I6 — CHECKED BY — Date
Value: 10/07/2023
Reason: Checked Date >= Analysed Date.

✓ COMPLIANT I7 — APPROVED BY — Name
Value: Ramakant Singh
Reason: Name present.

✓ COMPLIANT I8 — APPROVED BY — Designation
Value: QC. Manager
Reason: Designation matches requirement.

✓ COMPLIANT I9 — APPROVED BY — Date
Value: 10/07/2023
Reason: Approved Date matches Released Date and >= Checked Date.

Arithmetic Verification

Shelf life, date sequence, result vs spec limits, assay basis consistency, BET limit, signatory date order.

Check	Expected	Calculated	Difference	Status
SHELF LIFE CHECK	Exp. Date = Mfg. Date + 5 years	06/2023 + 5 years = 05/2028	0 months	✓ COMPLIANT
	⚠ Exp. Date is exactly 5 years after Mfg. Date.			
DATE SEQUENCE	Mfg. Date <= Sampling Date <= Released Date <= Exp. Date	06/2023 <= 07/07/2023 <= 10/07/2023 <= 05/2028	All dates in correct sequence	✓ COMPLIANT
	⚠ All dates are in correct chronological order.			
SIGNATORY DATES	Analysed <= Checked <= Approved	10/07/2023 <= 10/07/2023 <= 10/07/2023	All dates equal	✓ COMPLIANT
	⚠ All signatory dates are the same and in correct order.			
RESULT vs SPEC: Residue On Ignition	NMT 0.1%	0.14%	0.04% above limit	✗ NON-COMPLIANT
	⚠ 0.14% > 0.1% (NON_COMPLIANT)			
RESULT vs SPEC: Water Determination	NMT 2.0%	0.25%	Within limit	✓ COMPLIANT

	⚠ 0.25% < 2.0%			
RESULT vs SPEC: Assay	NLT 98.0% and NMT 102.0%	99.03%	Within range	✔ COMPLIANT
	⚠ 98.0% <= 99.03% <= 102.0%			
STERILITY/BET RELEASE WINDOW	Released Date ≤ Sampling Date + 14 days	10/07/2023 - 07/07/2023 = 3 days	3 days	✔ COMPLIANT
	⚠ Release within 14 days of sampling.			
PARAMETER SEQUENCE	COA parameter order matches API Spec	Order matches	No difference	✔ COMPLIANT
	⚠ COA and API Spec parameter order are identical.			

Cross-Document Mismatches

Values in the COA that differ from reference documents or trained pharmaceutical data.

Property	COA Value	Ref Doc	Ref Value	Trained Data	Impact
Residue On Ignition	0.14% w/w	DOC9(Spec-API)	NMT 0.1% w/w	NMT 0.1% (USP 44, BP Monograph)	Result exceeds specification; CRITICAL non-compliance.

Recommended Actions

CRITICAL

Action 1: Residue On Ignition result exceeds specification (0.14% vs NMT 0.1%).

Error in: **DOC1 (COA_API)** → "0.14% w/w"

Correct: "NMT 0.1% w/w"

Supporting: [object Object], [object Object]

Action: **CORRECT**

Inconclusive Resolutions



Document Needed: Batch manufacturing record or stability data

Reason: To verify batch size, manufacturing date, and sampled quantity.

Affected (3): A7, A8, A12

Reviewer Remarks

- The COA for Tetracaine Hydrochloride USP is generally well-aligned with reference specifications and pharmacopoeial standards.
- However, the Residue On Ignition result (0.14%) exceeds the NMT 0.1% limit per USP 44 and internal specification, representing a critical non-compliance.
- All other parameters, including assay, impurities, identity, and microbiological tests, are compliant.
- Immediate corrective action is required for the Residue On Ignition result before this batch can be considered for release or submission.